

Overview of Bagging ensemble

Subject: Bagging ensemble

1.

Description of the technique Given a standard training set D of size n , bagging generates m new training sets D_i , each of size n' , by sampling from D uniformly and with replacement. By sampling with replacement, some observations may be repeated in each D_i . If $n' = n$, then for large n the set D_i is expected to have the fraction $(1 - 1/e)$ (~63.2%) of the unique samples of D , the rest being duplicates. This kind of sample is known as a bootstrap sample. Sampling with replacement ensures each bootstrap is independent from its peers, as it does not depend on previous chosen samples when sampling. Then, m models are fitted using the above bootstrap samples and combined by averaging the output (for regression) or voting (for classification).

2.

Because of their properties, random forests are considered one of the most accurate data mining algorithms, are less likely to overfit their data, and run quickly and efficiently even for large datasets. They are primarily useful for classification as opposed to regression, which attempts to draw observed connections between statistical variables in a dataset. This makes random forests particularly useful in such fields as banking, healthcare, the stock market, and e-commerce where it is important to be able to predict future results based on past data. One of their applications would be as a useful tool for predicting cancer based on genetic factors, as seen in the above example. There are several important factors to consider when designing a random forest. If the trees in the random forests are too deep, overfitting can still occur due to over-specificity. If the forest is too large, the algorithm may become less efficient due to an increased runtime. Random forests also do not generally perform well when given sparse data with little variability. However, they still have numerous advantages over similar data classification algorithms such as neural networks, as they are much easier to interpret and generally require less data for training. As an integral component of random forests, bootstrap aggregating is very important to classification algorithms, and provides a critical element of variability that allows for increased accuracy when analyzing new data, as discussed below.

3.

Improving Random Forests and Bagging While the techniques described above utilize random forests and bagging (otherwise known as bootstrapping), there are certain techniques that can be used in order to improve their execution and voting time, their prediction accuracy, and their overall performance. The following are key steps in creating an efficient random forest:

4.

Many weak learners aggregated typically outperform a single learner over the entire set, and have less overfit Reduces variance in high-variance low-bias weak learner, which can improve efficiency (statistics) Can be performed in parallel, as each separate bootstrap can be processed on its own before aggregation. Disadvantages:

5.

For a weak learner with high bias, bagging will also carry high bias into its aggregate. Loss of interpretability of a model. Can be computationally expensive depending on the dataset.

6.

Creating the out-of-bag dataset The out-of-bag dataset represents the remaining people who were not in the bootstrap dataset. It can be calculated by taking the difference between the original and the bootstrap datasets. In this case, the remaining samples who were not selected are Emily, Jessie, George, Rachel, and Jamal. Keep in mind that since both datasets are sets, when taking the difference the duplicate names are ignored in the bootstrap dataset. The illustration below shows how the math is done:

7.

Random Forests The next part of the algorithm involves introducing yet another element of variability amongst the bootstrapped trees. In addition to each tree only examining a bootstrapped set of samples, only a small but consistent number of unique features are considered when ranking them as classifiers. This means that each tree only knows about the data pertaining to a small constant number of features, and a variable number of samples that is less than or equal to that of the original dataset. Consequently, the trees are more likely to return a wider array of answers, derived from more diverse knowledge. This results in a random forest, which possesses numerous benefits over a single decision tree generated without randomness. In a random forest, each tree "votes" on whether or not to classify a sample as positive based on its features. The sample is then classified based on majority vote. An example of this is given in the diagram below, where the four trees in a random forest vote on whether or not a patient with mutations A, B, F, and G has cancer. Since three out of four trees vote yes, the patient is then classified as cancer positive.

8.

Creation of Decision Trees The next step of the algorithm involves the generation of decision trees from the bootstrapped dataset. To achieve this, the process examines each gene/feature and determines for how many samples the feature's presence or absence yields a positive or negative result. This information is then used to compute a confusion matrix, which lists the true positives, false positives, true negatives, and false negatives of the feature when used as a classifier. These features are then ranked according to various classification metrics based on their confusion matrices. Some common metrics include estimate of positive correctness (calculated by subtracting false positives from true positives), measure of "goodness", and information gain. These features are then used to partition the samples into two sets: those that possess the top feature, and those that do not. The diagram below shows a decision tree of depth two being used to classify data. For example, a data point that exhibits Feature 1, but not Feature 2, will be given a "No". Another point that does not exhibit Feature 1, but does exhibit Feature 3, will be given a "Yes".

9.

Bootstrap aggregating, also called bagging (from bootstrap aggregating) or bootstrapping, is a machine learning (ML) ensemble meta-algorithm designed to improve the stability and accuracy of ML classification and regression algorithms. It also reduces variance and overfitting. Although it is usually

applied to decision tree methods, it can be used with any type of method. Bagging is a special case of the ensemble averaging approach.